ket versions of Schrödinger's equation (Eq. 4.10), we get

$$\frac{d}{dt}\langle\Psi(t)|\mathbf{L}|\Psi(\mathrm{t})\rangle = \frac{i}{\hbar}\langle\Psi(\mathrm{t})|\mathbf{HL}|\Psi(\mathrm{t})\rangle - \frac{i}{\hbar}\langle\Psi($$

or, more concisely,

$$\frac{d}{dt}\langle\Psi(t)|\mathbf{L}|\Psi(\mathrm{t})\rangle = \frac{i}{\hbar}\langle\Psi(\mathrm{t})|\ [\mathbf{HL}-\mathbf{LH}]\ |\Psi(\mathrm{t})\rangle. \qquad (4.16)$$

If you are used to ordinary algebra, Eq. 4.16 has a strange appearance. The right-hand side contains the combination **HL**− **LH**, a combination that would ordinarily be zero. But linear operators are not ordinary numbers: when they are multiplied (or applied sequentially), the order counts. In general, when **H** acts on **L**|Ψ>, the result is not the same as when **L** acts on **H**|Ψ>. In other words, except for special cases, **HL** ≠ **LH**. Given two operators or matrices, the combination

LM − **ML**

is called the *commutator* of **L** with **M**, and it is denoted by a special symbol:

LM − **ML** = [**L**, **M**].

It's worth noticing that [**L**, **M**] = −[**M**, **L**] for any pair of operators. Armed with the notation for commutators, we can now write Eq. 4.16 in a simple form:

$$\frac{d}{dt}\langle\mathbf{L}\rangle = \frac{i}{\hbar}\langle[\mathbf{H},\mathbf{L}]\rangle \qquad (4.17)$$

or, equivalently,